MIL-C-915/51B
5 April 1973
SUPERSEDING 1/
MIL-C-915/51A
1 August 1972

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE 3SU

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cable described herein shall consist of this document and the latest issue of Specification MIL-C-915.

REQUIREMENTS:

Qualification required.

(B)

Construction

First - Copper conductor, AWG No. 18 (7 X 0.0152 inch), uncoated. Maximum diameter 0.050 inch.
Second - Polyethylene insulation, type II, class L, grade 4 of L-P-390. Nominal wall thickness 0.020 inch. Colored insulation, one black, one white and one red conductor for each triad.
Third - Clear polyamide jacket on each conductor, type III, grade E or type IV of MIL-M-20693. Nominal thickness 0.003 inch. Nominal diameter 0.092 inch.
Fourth - Three conductors (one black, one white and one red) cabled together with a nominal lay of 3.5 inches to form each triad.
Fifth - Binder tape over each triad, at manufacturer's option, applied helically with overlap.
Sixth - Braided shield of AWG No. 34 or No. 36 uncoated copper. Braid angle 30 to 35 degrees; minimum coverage of 85 percent.
Seventh - Shield insulation of one polyester tape, type G of MIL-I-631, plus a jacket of clear polyamide, type III, grade E or type IV of MIL-M-20693. Nominal thickness 0.003 inch. Standard Identification Code, applied by Method 2.
- Alternate shield insulation of two polyester tapes, type G of MIL-I-631, sealed. Standard Identification Code, applied by Method 2, on inner tape.
- Diameter over completed triad 0.240 inch nominal.
Eighth - The required number of triads cabled with a lay not greater than 24 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with triad No. 1, from center outward. Fibrous fillers as necessary to form a firm, round cable.
Ninth - Binder tape applied helically with overlap.
Tenth - Standard thermoplastic jacket (gray). (See Details for thickness.) Cable surface marking required.

Details

Type and size	Number of triads	Cable jacket thick. Min. (inch)	Overall diameter	
			Min. (inches)	Max. (inches)
3SU-3	3	0.040	0.650	0.700
3SU-7	7	.040	.840	.910
3SU-10	10	.050	1.100	1.190
3SU-14	14	.050	1.200	1.290
3SU-19	19	.050	1.340	1.430

1/ Requirements formerly covered on MIL-C-24145/11.

(B) Denotes changes

FSC 6145

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Details (Cont'd)

Type and size	Number of triads	Cable jacket thick. Min. (inch)	Overall diameter	
			Min. (inches)	Max. (inches)
3SU-24	24	0.050	1.580	1.670
3SU-30	30	.050	1.680	1.770
3SU-37	37	.060	1.840	1.930
3SU-44	44	.060	2.060	2.150

EXAMINATION AND TESTS:

	Requirements:
Basic electrical:	
Conductor resistance - At 25°C., Ohms/1000 ft., maximum	6.80
Voltage withstand - Volts, rms, minimum	
Conductor to conductor	2000
Conductor to shield	1000
Shield to shield	500
Group A:	
Visual and dimensional	
Capacitance	
Mutual copacitance - pF/ft. at 1 kHz, maximum	30
Capacitance unbalance - Percent, maximum	5
Group B:	
Physicals (unaged)	
Insulation (conductor)	
Tensile strength - psi, minimum	1400
Elongation - Percent, minimum	300
Jacket (cable)	
Tensile strength - psi, minimum	1800
Elongation - Percent, minimum	225
Group C:	
Physicals (aged)	
Insulation (conductor)	
Air Oven	
Elongation - Percent of unaged, minimum	60
Jacket (cable)	
Hot oil immersion	
Tensile strength - psi, minimum	1400
Elongation - Percent, minimum	50
Permanence of printing (jacket) - Cycles, minimum	250

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical plus all of Groups A, B and C.

UNIT ORDERING LENGTHS:

Size	Feet (nominal)
3 through 19	1000
24 through 44	500

Custodians:
Army - MI
Navy - SH
Air Force - 80
Review activities:
Army - EL, MI, AV
Navy - EC
Air Force - 80
User activities:
Army - ME, MU, WC
Navy - CG

Preparing activity:
Navy - SH
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